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VMD-based iterative Boruta feature extraction and CNNA-BiLSTM for short-term load forecasting

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ABSTRACT

Short-term load forecasting is vital for efficient power system planning. To better extract features expanded by the signal decomposition method and enhance the performance of load prediction models, this paper proposes an iterative Boruta feature extraction method based on variational modal decomposition (VMD-IB) and a bi-directional long short-term memory network assisted by convolutional neural network and self-attention mechanism(CNNA-BiLSTM). Firstly, VMD decomposes the load data to capture different patterns and scales. An iterative Boruta method is proposed to identify the most valuable features for load prediction through an adaptive iterative strategy during feature selection. VMD-IB shows an advantage in extracting correlated features at multiple scales, enabling the model to better capture complex load fluctuations. Secondly, CNNA-BiLSTM enhances the ability of BiLSTM to fit the nonlinear data using CNN for temporal feature extraction and SAM for attention and weight allocation, improving the feature expressiveness and prediction performance. Lastly, extensive experiments on New York Independent System Operator(NYISO) data validate the framework. Compared to the state-of-the-art method, the proposed method achieves a 12% improvement in mean absolute percentage error and an 8% improvement in the Theil inequality coefficient, showcasing the superiority of the proposed framework in accurately predicting short-term load demand.

1. Introduction

Electricity load forecasting is a critical task that relies on an in-depth analysis of historical data and a combination of factors. This enables the generation of dependable forecasts for future load demand, ensuring accurate predictions for electricity planning and management [1]. Accurate load forecasting provides utilities with the opportunity to achieve substantial cost savings [2]. With the rapid advancement of science and technology, the use of electric power is increasingly widespread. Therefore, the fluctuation of power load is affected by more and more factors and shows higher levels of non-stationarity and complexity, posing more significant challenges for short-term power load forecasting [3].

The current load forecasting methods can be classified into two categories: traditional statistical forecasting methods and modern intelligent methods. Traditional statistical forecasting methods include linear regression (LR) [4], autoregressive moving average (ARMA) [5,6], and autoregressive integrated moving average (ARIMA) [7–9]. Although these methods provide simple models and fast prediction speeds, most of them rely on linear relational models, making it challenging to improve prediction accuracy for electric loads with high nonlinearity and non-stationarity. With the advancement of artificial intelligence, modern intelligent methods can extract potential information from large volumes of data and automatically establish general prediction models. For instance, support vector machines (SVM) are used to process continuously generated data from smart grids in real time, which helps in timely and accurate forecasting of power loads [10]. A short-term load forecasting method based on SVM and k-means clustering is proposed, which utilizes similar days as input data to achieve improved accuracy compared with traditional methods [11]. An enhanced weighted transfer learning algorithm based on TrAdaBoost is proposed for daily peak load forecasting during holidays, yielding favorable prediction results [12]. Moreover, the multi-objective locust optimization algorithm (MOGOA) is integrated into the parameter optimization of artificial neural networks, achieving high accuracy and robustness [13]. These modern intelligent methods have gained extensive application in time series analysis due to their superior fitting capabilities for nonlinear sequences.

To further improve the accuracy of load forecasting, researchers have employed signal decomposition methods in the preprocessing of load sequences [14]. One approach involves modeling the decomposed

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sequences individually for prediction and combining the results. Liang et al. [15] recommend the utilization of empirical mode decomposition (EMD) to decompose the original load sequence into intrinsic mode functions (IMFs) and residuals with distinct frequencies. Then, the decomposed components are utilized for load forecasting through GRNN models. Fan et al. [16] decompose the historical load data into high-frequency, random, and low-frequency components using ensemble empirical mode decomposition (EEMD) and utilizes a hybrid model for load prediction. Experimental results demonstrated that the proposed model outperformed other models. He et al. [17] combine variational modal decomposition (VMD) and long short term memory (LSTM) networks for load forecasting, which fully uses the time and frequency domain characteristics of load demand. While the utilization of decomposition algorithms positively impacts load forecasting to a certain extent, it unavoidably increases the complexity of prediction models and introduces random cumulative errors. Another approach involves utilizing them as input features, providing richer signal trend features. Liu et al. [18] utilize EMD for load decomposition and selects subsequences highly correlated with the original load data as input features. Load forecasting is conducted using the 1D-CNN-TCN-LSTM model. Nevertheless, the EMD algorithm suffers from mode mixing, which can lead to the blending of different components. Therefore, when selecting a decomposition algorithm, it is essential to consider the significance of the information in the resulting load components and choose an appropriate decomposition algorithm.

Using signal decomposition to incorporate sub-sequences as features increases the dimensionality of the feature space. On the one hand, this method could improve the prediction accuracy. On the other hand, it may also introduce irrelevant and redundant features, resulting in increased training time and reduced prediction accuracy of the models [19]. Therefore, it is necessary to conduct feature selection before training. Previous studies use the Pearson correlation coefficient (PCC) to calculate the correlation between influencing factors and load, e.g., meteorological factors, date type, and economy. The key influencing factors are then selected for load forecasting to reduce dimensionality, which improves the model's efficiency [20,21]. In [22], essential weather variables selected using Pearson and Spearman correlation techniques are dynamically incorporated to predict the day-ahead and week-ahead load demand, showing promising prospects of this method. Lu et al. [23] use the maximum information coefficient (MIC) to select data for use in LSTM to improve the accuracy of load forecasting in case of scarcity of historical data. Yet, these methods still have significant limitations due to the complex nonlinear relationship between features and load data. To overcome this problem, shao et al. [24] employs Shapley values to select the optimal feature set for prediction and evaluate each feature's contribution. However, this method requires manual threshold setting and a lack of flexibility. The Boruta algorithm overcomes this limitation. In [25], the Boruta feature selection based on a wrapper method is employed to address the curse of dimensionality and overfitting problems, and the bidirectional long short-term memory (Bi-LSTM) model is used to improve wind speed prediction performance. The Boruta algorithm shows considerable advantages in searching for all relevant features. Regardless, it is sensitive to the classification performance of unimportant features, which may lead to excluding some helpful features for load forecasting and impose certain limitations on model learning.

Deep learning methods have rapidly developed and found wide applications in electric load forecasting. For instance, atef et al. [26] use a Bi-LSTM network for training and prediction, demonstrating the effectiveness of the proposed short-term load forecasting method. Dong et al. [27] utilize a spatiotemporal convolutional network (STCN) with a directed graph convolutional structure to capture the temporal features of wind power, further improving wind power prediction performance. Wang et al. [28] propose a model based on the combined effects of weather and modified temperature-ElmanNN, which effectively captures the effects of various meteorological factors and improves forecasting accuracy. Still, individual models have limitations in providing comprehensive data interpretation and modeling capabilities, failing to capture broader information and features. To address this issue, kim et al. [29] combine CNN and LSTM networks to extract spatial and temporal features for predicting housing energy consumption, achieving improved prediction performance. In [30], the proposed model combines the temporal convolutional network (TCN) channel and the temporal attention mechanism (AM), which makes full use of the nonlinear relationship between meteorological factors and loads, and thus effectively improves the accuracy and generalization ability of load forecasting. However, the methods mentioned above still have some limitations regarding feature extraction and utilization for time series data, which indicates the potential for further research in deep learning methods.

To tackle the challenge of feature extraction after expanding them through signal decomposition methods and to enhance the ability of deep learning in feature extraction and augmentation, this paper proposes an iterative Boruta feature extraction method based on variational modal decomposition (VMD-IB) and a bi-directional long short-term memory network assisted by convolutional neural network and self-attention mechanism (CNNA-BiLSTM). The main contributions of the paper can be described as follows.

(1) The load sequences are decomposed using variational modal decomposition (VMD) to extract the underlying trend information. This decomposition yields multiple components that serve as candidate features, reflecting the fluctuation patterns of the load sequences across different periods and patterns. To effectively extract informative features from these candidate features, which have been augmented with VMD load components, an iterative Boruta (IB) method is employed for adaptive feature selection, which effectively reduces the complexity of model learning and improves convergence speed.

(2) CNNA-BiLSTM model is developed by integrating convolutional neural network (CNN) and self-attention mechanism (SAM) modules into the bi-directional long short-term memory (BiLSTM) model. This integration enhances the model's feature extraction capability and leverages the utilization of crucial information. The CNNA-BiLSTM model effectively captures intricate relationships and dependencies in the data.

(3) Extensive experiments on publicly available datasets from the New York Independent System Operator (NYISO) are conducted to evaluate the effectiveness and performance of the proposed method. The results show the superiority in short-term load forecasting compared with the state-of-the-art method.

The subsequent sections of this paper are structured as follows: Section 2 outlines the proposed model framework for prediction. Section 3 delivers data description and data preprocessing. Section 4 presents the experimental design and analysis conducted in this paper. Finally, Section 5 provides a comprehensive summary of the research.

2. Method

This section provides an overview of the fundamental theories behind the proposed VMD-IB and CNNA-BiLSTM framework. We will discuss the key concepts and principles of VMD, Boruta, CNN, SAM, and BiLSTM, which form the foundation of our proposed methodology.

2.1. VMD-based load component feature module

2.1.1. VMD

VMD is a well-established signal decomposition technique with a robust theoretical foundation. In contrast to other methods like EMD and EEMD, VMD effectively addresses mode mixing and endpoint effects commonly encountered during decomposition. This makes it particularly suitable for handling load sequences that show smooth variations. A significant portion of the noise can be attenuated by employing VMD, reducing signal non-stationarity. The decomposition

process yields multiple groups of IMFs. Let the original load sequence to be processed be denoted as $f(t)$. The constraint expression for signal decomposition using VMD can be defined as follows:

$$\min_{\{u_k,w_k\}} \left\{ \sum_k \left\| \partial_t \left[(\delta(t) + j/\pi t) * u_k(t) \right] e^{-jw_k t} \right\|_2^2 \right\}$$
$$\text{s.t.} \sum_{k=1}^{N} u_k = f(t) \tag{1}$$

In the above formula: u_k and w_k represent the kth modal component and the center frequency after decomposition, respectively. The parameter N indicates the number of modal decompositions and $\delta(t)$ denotes the Dirac function. ∂_t denotes the partial derivation. The symbol $*$ represents the convolution operator.

The specific process of VMD method decomposition is as follows:

Step 1: Initialize $\{u_k^1\}$, $\{w_k^1\}$, λ_k^1 and $n = 0$;

Step 2: $n = n + 1$, enter the cycle;

Step 3: The update formulas for u_k and w_k are iteratively applied until the number of decompositions reaches N, at which point the inner loop terminates.

Step 4: The updating formula for λ is applied to update λ iteratively.

Step 5: Given a tolerance ε, the loop is terminated if the stopping condition (2) is satisfied.

$$\sum_k \|u_k^{n+1} - u_k^n\|_2^2 / \left\|u_k^n\right\|_2^2 < \varepsilon \tag{2}$$

Otherwise, the loop continues with Step 2.

In the above process, u_k represents the decomposed single-component amplitude-modulated frequency-modulated (AM-FM) signal, w_k represents the center frequency of each AM-FM signal component, λ denotes the Lagrange multiplier, and n denotes the number of iterations.

2.1.2. VMD-based load component feature module

Deep learning models can capture complex patterns in time-series data by representing them as high-dimensional vectors. Nonetheless, these models can become overly reliant on the original historical data, leading to overfitting and limited extraction of significant features [31]. To overcome this constraint, some decomposition methods are proposed to divide the time series signal into sub-signals of varying scales. This decomposition yields a more comprehensive representation of the evolving patterns in power load, thereby augmenting the model's capacity to capture and comprehend the inherent dynamics.

The flowchart illustrating the load component feature module is depicted in Fig. 1. This module leverages the VMD algorithm to decompose the historical load into multiple intrinsic mode function components. These components and the original features are subsequently incorporated as candidate input features for the model. By integrating the decomposed components into the feature set, the model benefits from additional information that enhances its understanding of the data, resulting in improved prediction accuracy.

2.2. Iterative Boruta feature extraction module

2.2.1. Boruta

The Boruta method shares a similar concept with the random forest algorithm, employing randomness and gathering prediction outcomes to identify and eliminate less relevant features through statistical tests. This process aims to identify variables with significant correlations and discover new features.

The steps of the Boruta method:

Step 1: Build an extended feature set based on the initial feature set. The expanded feature set is:

$$F = \left\{ f_i \cup f_i', i = 1, 2, 3, \ldots, p \right\} \tag{3}$$

Among them, f_i represents the initial feature (feature) variable; f_i' represents the shadow feature variable; i represents the dimension of the feature variable.

Step 2: Implement the shuffle process on the expanded feature set. Randomly sort the original features and shadow features, obtain a random shadow feature set, and eliminate the correlation between the original features and each shadow feature.

Step 3: Use the random forest classifier to calculate the importance score set of all features for the extended feature set. That is, calculate the score index z-score of all features.

A Random Forest is being constructed as $\{h_1(F), h_2(F), h_3(F), \ldots, h_M(F)\}$, where M denotes the number of iterations of the Random Forest.

Compute the importance scores in terms of z-scores for variable features: $\{Z_1(F), Z_2(F), Z_3(F), \ldots, Z_M(F)\}$. There are M importance scores for each variable in the original features and each variable in the shadow features.

Step 4: The importance of the original feature variables is compared and tested, and the importance counts are labeled. Every time the random forest classifier is executed, it calculates the maximum value $MZSA$ of the shadow feature score (z-score):

$$MZSA = \max Z_j \left(f_i' \right) \tag{4}$$

Here, $i = 1, 2, 3, \ldots, p$; $j = 1, 2, 3, \ldots, M$

Test each original feature variable in turn. Initial hypothesis $H0$: $Z_j \left(f_i' \right) > MZSA$. Suppose the raw feature variable importance passes the test. In that case, i.e., the raw feature variable importance score is greater than the maximum of the shadow feature importance scores, and the feature is labeled with one importance count.

Step 5: Compare and determine the importance of the original feature variables. As previous studies have found, the number of importance markings (hits) for the original feature variables follows a Bernoulli distribution pattern:

$$hits \sim N(0.5N, \sqrt{0.25M}) \tag{5}$$

When the value of the original feature is significantly greater than the expected value, it is considered essential and is retained. When the value of the original feature is significantly smaller than the expected value, it is considered unimportant and is removed. However, the assessment of feature importance is influenced by the randomness inherent in the random forest classifier, making the evaluation results highly sensitive to the classification performance of unimportant features.

2.2.2. Iterative Boruta feature extraction module

The inclusion of load component features leads to an increase in the dimensionality of the input data. Nevertheless, abundant input features can challenge the model's learning process, and therefore a more effective method for feature selection is needed. The Boruta, which adopts the wrapper method of the random forest classification algorithm, integrates feature selection within the model construction process. This algorithm effectively captures the intricate nonlinear relationship between features and load data. However, the Boruta can be sensitive to the classification performance of unimportant features, resulting in a limited number of selected features. This limitation may impede the learning performance of the load forecasting model and potentially ignore certain valuable features for load forecasting.

This paper proposes an Iterative Boruta (IB) feature selection method for embedded feature selection to address this problem, as depicted in Fig. 2. The IB incorporates the decomposition method to characterize candidate features by extracting load components. It produces two outcomes: selected features (Group 1) and unselected features. Subsequently, the Boruta method is applied to the unselected features to identify newly selected and unselected features. The newly selected features are added to Group 1 (now Group 2), and the Boruta method is iteratively conducted on the remaining unselected features until either no unselected features remain or the number of features selected in the next round is the same as in the previous round. This iterative process completes the feature selection, resulting in the generation of m Groups of features. These feature groups are then individually utilized for model training, and the best results are selected based on the validation set.

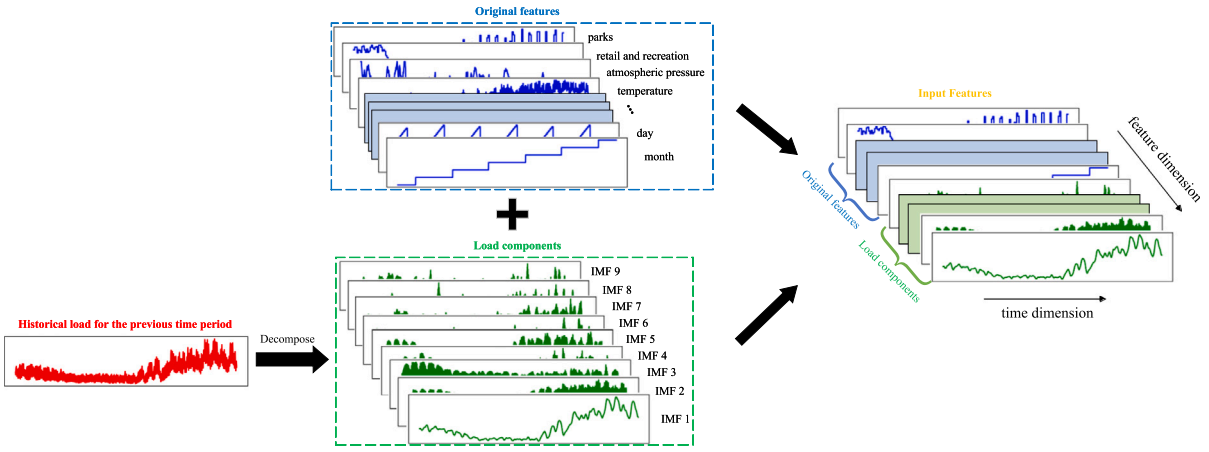


Fig. 1. VMD-based load component feature module.

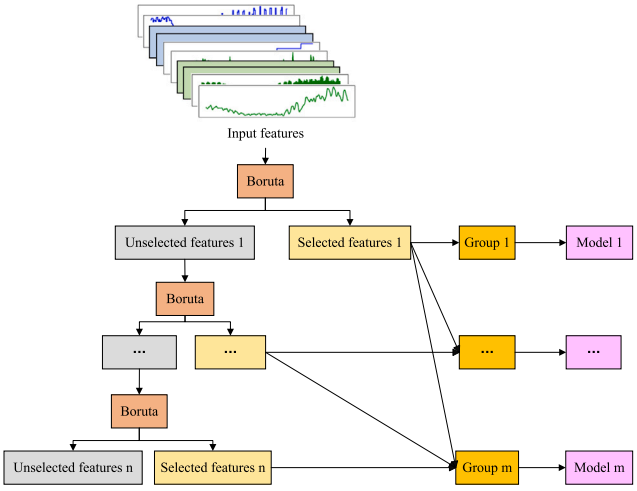


Fig. 2. Iterative Boruta feature extraction module.

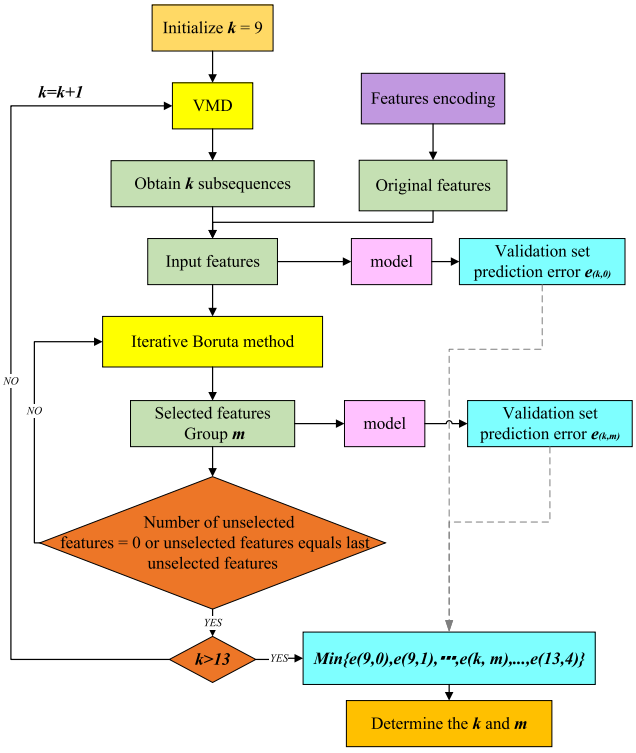


Fig. 3. The process of the VMD-IB framework.

2.3. VMD-IB framework

VMD-IB is a method that combines VMD-based load component feature and iterative Boruta feature selection. Initially, the VMD method is employed to decompose the load data, transforming the original load signal into multiple IMFs. Afterwards, the iterative Boruta feature extraction method is applied. Compared with traditional feature extraction methods in the time and frequency domains, VMD-IB provides a more comprehensive representation of the dynamic features of the load sequence. By combining the advantages of VMD and IB, the VMD-IB method establishes a robust framework for feature extraction and selection in load forecasting. The flowchart of the VMD-IB framework is illustrated in Fig. 3.

In this method, two crucial parameters must be determined: the number of subsequences generated by the VMD, denoted as k, and the number of feature sets adaptively selected by the IB, denoted as m. Multiple datasets are obtained by iterating through a two-layer loop, resulting in various datasets with corresponding errors $\{e(9,0), e(9,1), e(9,2), \ldots, e(k,m), \ldots, e(13,4)\}$ during the process of model training. Considering these errors, the dataset with the minimum error can be selected, allowing us to determine the most suitable values of k and m for completing the dataset construction process.

2.4. CNNA-BiLSTM framework

2.4.1. CNN

CNN is a specialized deep-feedforward network widely employed in various tasks, including image and natural language processing. The network's architecture is characterized by local connections and weight sharing, enabling it to capture spatial and temporal patterns to a certain extent. In the field of time series forecasting, researchers have extended the CNN framework and introduced a one-dimensional convolutional neural network (1DCNN) to address the unique challenges of predicting time series data [32].

In this architecture, the input data is typically represented as a one-dimensional or two-dimensional array representing the time series. The 1DCNN applies convolutional kernels in the temporal dimension of the sequence, sliding them across the input to extract relevant features. Fig. 4 illustrates the feature extraction process from the input sequence

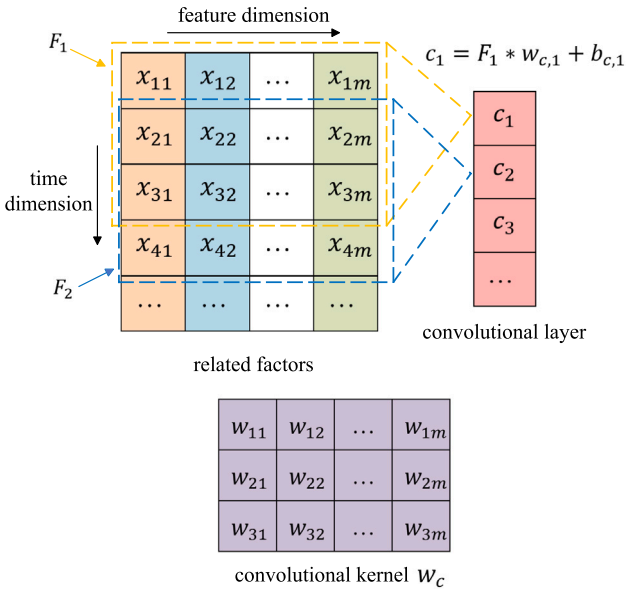


Fig. 4. 1DCNN convolution feature extraction process.

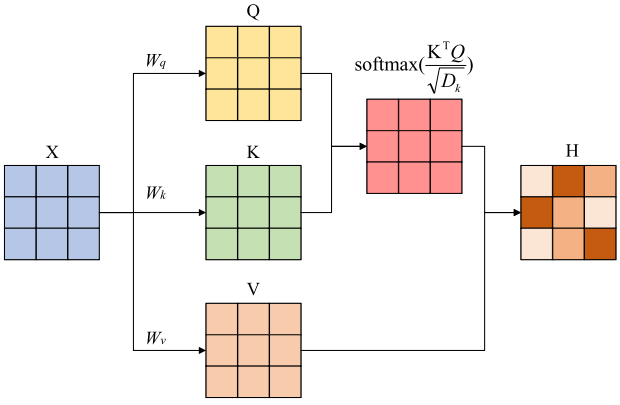


Fig. 5. Calculation process of SAM.

using 1DCNN. As the convolution operation is applied along the time dimension, the network captures inter-temporal correlations and aligns the input sequence with the relevant factors. The convolution kernel can be conceptually viewed as a sliding time window, allowing the network to capture the input data's temporal dynamics effectively.

2.4.2. SAM

When dealing with large-scale sequence data using neural networks, it is beneficial to incorporate attention mechanisms, which is inspired by the human brain. Attention mechanisms selectively focus on key information while disregarding less relevant details, improving network performance. Self-attention models enable the dynamic generation of weights and can effectively handle sequences of varying lengths, establishing long-range dependencies. The computational process is illustrated in Fig. 5.

Given an input sequence $X = [x_1, \ldots, x_n]$ and an output sequence $H = [h_1, \ldots, h_n]$, each input x_i is linearly mapped to three distinct spaces, yielding query vectors q_i, key vectors k_i, and value vectors v_i. The mapping computation is performed as follows:

$$\begin{aligned} Q &= W_q X \\ K &= W_k X \\ Q &= W_q X \end{aligned} \tag{6}$$

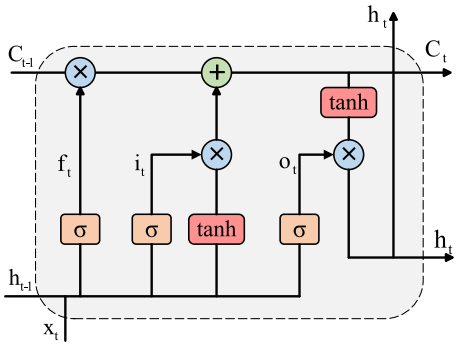


Fig. 6. LSTM hidden layer network structure.

where W_q, W_k, W_v represent the weight matrices for linear mappings, Q, K, V denote the query matrix, key matrix, and value matrix, respectively.

The calculation process of the output vector sequence is as follows:

$$H = V \operatorname{softmax}\left(\frac{K^T Q}{\sqrt{D_k}}\right) \tag{7}$$

where D_k represents the dimensionality of the matrices and $softmax$ refers to the function that performs column-wise normalization.

2.4.3. BiLSTM

LSTM networks are a specific recurrent neural network (RNN) architecture type. By effectively modifying the hidden layer's forgetting gate, input gate, and output gate, LSTM can capture long-term dependencies in the input sequence, overcoming the problem of regular RNNs' limited ability to learn long-range temporal dependencies. The hidden layer structure of the LSTM network is illustrated in Fig. 6.

The process of information transfer update is:

$$i_t = \sigma\left(W_{xi} x_t + W_{hi} h_{t-1} + W_{ci} c_{t-1} + b_i\right) \tag{8}$$

$$f_t = \sigma\left(W_{xf} x_t + W_{hf} h_{t-1} + W_{cf} c_{t-1} + b_f\right) \tag{9}$$

$$c_t = f_t c_{t-1} + i_t \cdot \tanh\left(W_{xc} x_t + W_{hc} h_{t-1} + b_c\right) \tag{10}$$

$$o_t = \sigma\left(W_{xo} x_t + W_{ho} h_{t-1} + W_{co} c_{t-1} + b_o\right) \tag{11}$$

$$h_t = o_t \cdot \tanh\left(c_t\right) \tag{12}$$

In the LSTM network, the input at time t, denoted as x_t, flows into the network through the input gate i_t, forget gate f_t, and output gate o_t, which determine the information flow within the network. The links weights for the recurrent layer are represented by $W_h i$, $W_h f$, $W_h o$ and $W_h c$ while the weights for the input layer are denoted as $W_x i$, $W_x f$, $W_x o$ and $W_x c$. The biases of the network are represented by b_i, b_f, b_o and b_c. The sigmoid activation function σ is applied, and the output quantity is denoted as h_t.

The bi-directional long short-term memory (BiLSTM) combines forward LSTM and backward LSTM. Compared with a single LSTM network, BiLSTM can capture both forward and backward relationships in the input data sequence by reusing network weights, which allows it to fully utilize its training capability without requiring excessive data and effectively reduces the risk of underfitting.

2.4.4. CNNA-BiLSTM framework

The CNNA-BiLSTM model consists of several crucial components, including the convolutional layer, self-attention mechanism, bidirectional long short-term memory network, and a fully connected layer. Fig. 7 illustrates the overall architecture of CNNA-BiLSTM.

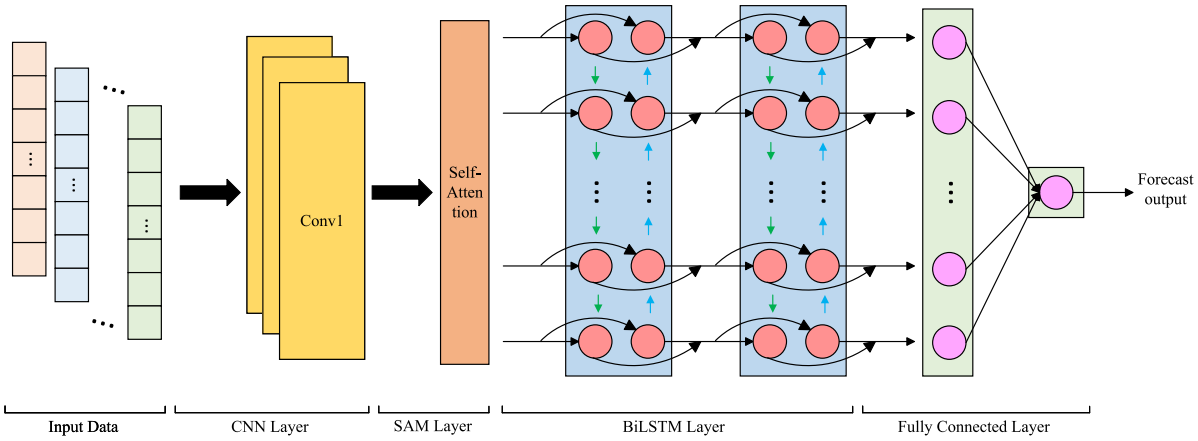


Fig. 7. CNNA-BiLSTM framework.

The convolutional layer extracts features from the input data, facilitating time series analysis. It applies filters to capture relevant patterns and extract important features from the input sequence. The self-attention mechanism is incorporated to prioritize and focus on key information. By assigning appropriate attention weights, the model can emphasize important features and enhance its ability to capture crucial data patterns. Subsequently, the BiLSTM network is employed to capture long-term dependencies in the data. By considering both the forward and backward relations of the input data sequence during training, the model can better capture the temporal dependencies and improve the overall fitting ability of the extracted features. Finally, the extracted features are forwarded to the fully connected layer, where the final output is generated. The fully connected layer processes the information from the preceding layers and produces the ultimate prediction or classification results.

The CNNA-BiLSTM model integrates the capabilities of CNN, SAM, and BiLSTM networks to improve prediction performance. Combining these three components makes the model more adept at capturing intricate patterns, leading to improved prediction accuracy.

2.5. Overall framework

The procedure of this proposed framework is as follows:

Step 1: Apply VMD on the preprocessed load data to extract component features.

Step 2: Combine the acquired component features with the original ones, thus creating a new feature set.

Step 3: Use IB for feature extraction, identifying the important features for load forecasting and refining the feature set.

Step 4: Input the optimized feature set into the model, passing through CNN, SAM, two-layer BiLSTM, and finally through a fully connected layer, to output load prediction results.

3. Data description and data preprocessing

3.1. Construction of the dataset

The experimental design and analysis are based on the data obtained from NYISO.[1] The data collection interval is one hour, spanning from February 15, 2020, to August 15, 2021, resulting in 13,152 data points. Additional data are also collected, including holiday information, meteorological factors,[2] and COVID-19 related data from Google's COVID-19 page.[3] These data sources are summarized in Table 1. The missing values are filled with the average of the neighboring data points to ensure the integrity of the dataset.

Table 1
Acquired data.

Category	Number of features	Feature contents
Time	5	Month, Day, Week, Hours, Holiday
Meteorology	6	Temperature, Atmospheric pressure, Relative Humidity, Average wind speed, Visibility, Dewpoint temperature
Mobility	6	Retail and recreation, Grocery and pharmacy, Parks, Transit stations, Workplaces, Residential

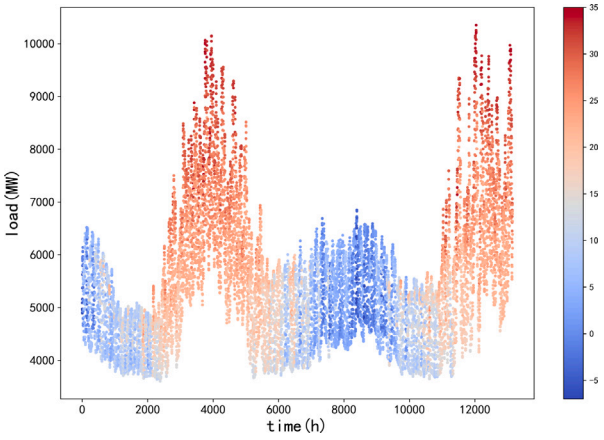


Fig. 8. The relationship between temperature and load.

3.2. Selection and analysis of relevant factors

The temperature at the same time scale as the load is the main focus of investigating the correlation between load and weather conditions. A scatter plot is created to visualize the relationship between temperature and load, as illustrated in Fig. 8. The load values are represented on the left side of the y-axis, while the temperature scale is shown on the right side. The figure shows that the load has a specific periodicity, in that the load fluctuations are regular. Moreover in the red region, as the temperature increases, the load value also increases. It shows that there is a correlation between temperature and load.

[1] https://www.nyiso.com.
[2] https://rp5.ru/Weather_in_New_York,_USA.
[3] https://www.google.com/COVID19/Mobility/.

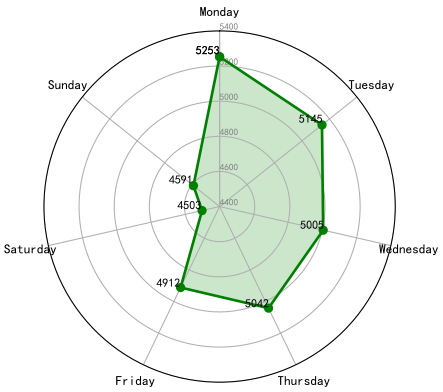


Fig. 9. Changes in load values in one week.

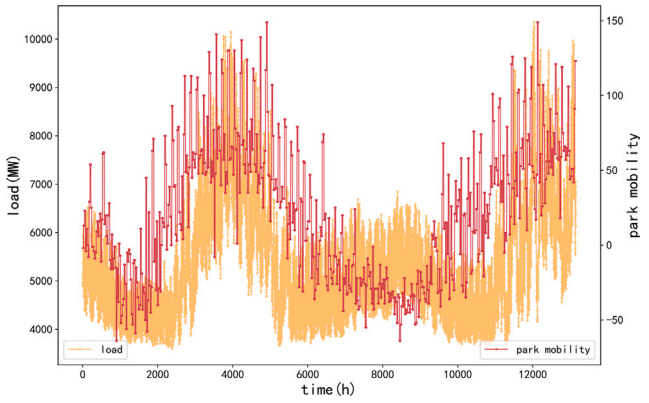


Fig. 10. The relationship between park mobility and load.

The load data for a week is extracted, and the daily load averages are calculated. These averages are then used to create a radar chart, as depicted in Fig. 9. The load demand shows a clear regularity, with lower loads on weekends compared with weekdays. Furthermore, load patterns differ during weekends, holidays, and special festivals, leading to load fluctuation.

In the study presented in [33], mobility data was introduced for load forecasting, and the relationship between mobility data and load was analyzed in detail. The study concluded that there is a significant correlation between mobility data and load, and incorporating mobility data improves forecasting accuracy. For instance, park mobility data was selected as a representative type of mobility data and analyzed alongside load data. Fig. 10 shows the relationship between load data (orange curve) and park mobility data (red curve). Observations show that there is a synchronized trend between the two during peak and trough periods: park mobility data tends to increase during peak load periods and decreases during trough periods. This suggests an unignorable correlation between park mobility data and load data.

This paper incorporates historical load data, time, holidays, meteorological factors, population mobility indicators, and load component indicators as input matrices. The load forecasting model in this paper is shown in the formula:

$$y_t = f\left(y_{t-1}, y_{t-2}, \ldots, y_{t-72}, o, d, w, h, type, p_t, s_t, IMF_1, IMF_2, \ldots, IMF_k\right) \tag{13}$$

where y_t represents the load prediction value at time t, $f(\cdot)$ denotes the non-linear mapping between external features and load prediction, $y_{t-1}, y_{t-2}, \ldots, y_{t-72}$ are the load values of the past 72 h steps. $o \in [1, 2, \ldots, 12]$ represents the 12 months in a year, $d \in [1, 2, \ldots, 31]$ represents the 31 days in a month, $w \in [1, 2, \ldots, 7]$ represents the 7 days in a week, $h \in [1, 2, \ldots, 24]$ represents the 24 h in a day, $type \in [0, 1]$ indicates holidays including weekends, p_t and s_t represent meteorological and population mobility features related to power load, and $IMF_1, IMF_2, \ldots, IMF_k$ represent k sub-sequences.

3.3. Evaluation metrics

Before proceeding with the experimentation, it is essential to establish evaluation metrics to evaluate the prediction accuracy. This paper employs mean absolute percentage error (MAPE) and Theil inequality coefficient (TIC) as evaluation metrics to measure the error between the predicted and actual values. The actual load values are used as the reference baseline for comparison. Let there be n samples, with the actual load value denoted as y and the predicted load value by the model denoted as y'. The error between them can be represented using the following formulas:

Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i' - y_i}{y_i}\right| \times 100\% \tag{14}$$

Theil Inequality Coefficient (TIC):

$$TIC = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i' - y_i\right)^2}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n} y_i'^2}\sqrt{\frac{1}{n}\sum_{t=1}^{N} y_i^2}} \tag{15}$$

The smaller the values of TIC and MAPE, the better the predictive performance of the model.

3.4. Setting of model hyperparameters

For building and training the model, we used the Keras library in Python and the Tensorflow backend. The model consists of one convolutional layer and two BiLSTM layers, where the CNN convolutional layer has 32 filters, a kernel size of 3, and uses the ReLU activation function. The first BiLSTM layer has 24 neurons, and the second BiLSTM layer has 48 neurons, with an integrated attention mechanism. Following the BiLSTM layers, we applied a fully connected layer with 24 and 1 neurons for output. After multiple trials and adjustments, we set the training epochs to 100, batch size to 24, the loss function to mean squared error, the evaluation metric to mean absolute percentage error, and the optimizer to Adam.

4. Experimental design and analysis

In this section, a series of experiments are conducted on publicly available datasets to evaluate the effectiveness of the proposed method. All models are implemented in Python 3.7, running on a PC equipped with a 3.7 GHz Intel i9-10900X CPU, 24 GB memory, and NVIDIA RTX 3090 GPU. The deep learning framework used is TensorFlow 2.6, with Keras 2.6, and the operating system is Windows 10 64-bit.

4.1. Case 1: Effectiveness of VMD-based load component feature module

To validate the effectiveness of the VMD-based load component feature module, a dataset covering the period from February 15, 2020, to August 15, 2020, is used. Six months of data are used to predict the load of the upcoming week. The training set comprises data from February 15 to August 1, the validation set encompasses data from August 2 to August 8, and the test set consists of data from August 9 to August 15. Consistently throughout our experiments, the CNNA-BiLSTM model is employed for training purposes.

We conduct a comparative experiment with EMD, EEMD, and CEEMDAN in a specific case. EMD, EEMD, and CEEMDAN are adaptive algorithms that allow decomposition without manual parameter

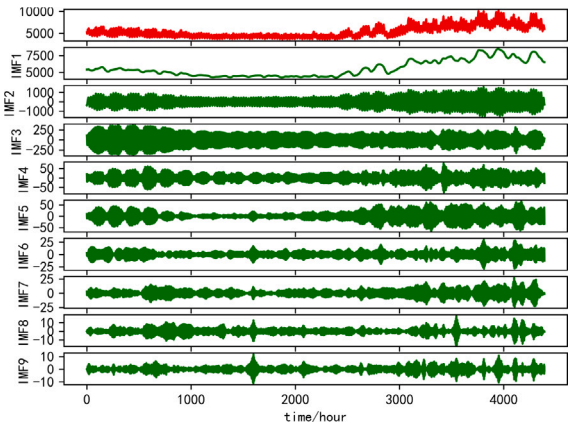


Fig. 11. Decomposed sub-sequences using VMD.

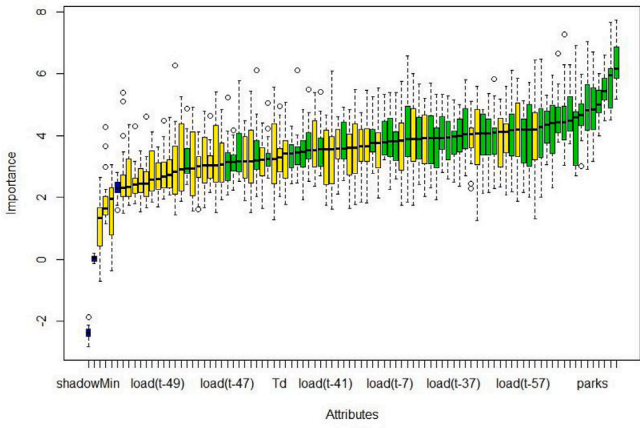


Fig. 12. Feature selection using Boruta.

Table 2
Maximum value of Pearson correlation coefficient among components.

Method	EMD	EEMD	CEEMDAN	VMD
Max PCC	0.829	0.826	0.438	**0.883**

Table 3
Prediction results with added load components.

Load components	Dim	Validation set		Test set	
		MAPE (%)	TIC × 100	MAPE (%)	TIC × 100
Original	89	1.05	0.687	0.967	0.622
EMD	97	0.99	0.661	0.881	0.543
EEMD	100	0.885	0.566	0.903	0.53
CEEMDAN	96	1.026	0.704	0.841	0.599
VMD	98	**0.641**	**0.405**	**0.55**	**0.339**

settings. In contrast, VMD requires manual input of the number of decompositions. In this experiment, the parameter k is defined as 9. Fig. 11 illustrates the subsequences obtained by decomposing the load sequence using VMD.

This paper employs the Pearson correlation coefficient for the calculation to analyze the correlation between sequence data and subsequences. For two sets of random variables X and Y, their correlation coefficient can be defined as follows:

$$\rho_{X,Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E\left(X^2\right) - E^2(X)}\sqrt{E\left(Y^2\right) - E^2(Y)}} \tag{16}$$

where E represents the expectation of random variables, the maximum value of the Pearson correlation coefficient among the components is computed, as high-frequency components have a minor impact on load forecasting. The results, shown in Table 2, indicate that among the sub-sequences obtained from the four decomposition algorithms for the dataset, the components generated by VMD show the highest correlation with the load, followed by EMD and EEMD, while CEEMDAN shows the lowest correlation.

Additional subsequences are incorporated with the original features to increase the dimensionality of the input features. Data normalization is performed to mitigate the influence of magnitude differences among the indicators and improve the model's convergence speed, mapping the values to the range [0, 1]. During this step, the model's load components obtained from EMD, EEMD, CEEMDAN, and VMD are included as new features, and the prediction results are compared. The outcome of this comparison is presented in Table 3.

Including the subsequence obtained from VMD demonstrates the best prediction performance for the validation and test sets compared to using only the original features. This finding is consistent with the results shown in Table 3, further confirming the VMD-based load component's significant contribution in load prediction. Notably, in the test set, all the features augmented with the subsequence show improved prediction accuracy compared to using only the original features. Particularly noteworthy is the VMD algorithm, which demonstrates the most substantial improvement in prediction accuracy, with a remarkable 43% improvement in MAPE and a notable 45% improvement in TIC.

Table 4
Prediction results of ib feature selection.

Feature selection	Dim	Validation set		Test set	
		MAPE (%)	TIC × 100	MAPE (%)	TIC × 100
Original	89	1.05	0.687	0.967	0.622
Original Group1	45	1.505	0.907	1.42	0.863
Original Group2	79	1.221	1.009	**0.796**	0.583
Original Group3	88	1.176	0.694	0.905	0.525
Original Group4	89	**0.927**	**0.634**	0.879	**0.569**

4.2. Case 2: Effectiveness of iterative Boruta feature extraction module

The data, models, and evaluation indicators used in this experiment are the same with Case 1. In the data processing stage, the Boruta method is used for feature selection, and the results are presented in Fig. 12. The x-axis represents the input features, while the y-axis represents the feature importance. As the horizontal axis increases, the importance of the features gradually increases, indicating that their influence gradually increases on the model. The green sections, most of which reside on the right side, represent features that were selected by Boruta, meaning that they had a significant impact in the load forecast. Conversely, others are unselected features that may show lower importance scores with small or insignificant contributions to load forecasting. The Boruta automatically eliminates these irrelevant features during the feature selection process to select the set of essential features.

Experiments are conducted on the original features to validate the effectiveness of the iterative Boruta feature extraction module. The features are inputted into the model in batches for training using the iterative Boruta method, and the results are presented in Table 4.

Group4 achieved the best or second-best results on both the validation and test sets, indicating that the feature selection of Group4 positively impacts the model's predictive performance. Despite having the same number of features as the original set, adjusting the order of input features still influenced the prediction results. This variation may be attributed to the differences in features obtained during the convolutional extraction of the data. However, the features selected by the iterative Boruta method showed greater robustness.

Furthermore, it can be observed that using the Boruta to select features only once did not yield satisfactory training results, suggesting

Table 5
Results of number of vmd-based load component features.

Number of components	Dim	Validation set		Test set	
		MAPE (%)	TIC × 100	MAPE (%)	TIC × 100
Original	89	1.05	0.687	0.967	0.622
VMD9	98	0.641	0.405	0.55	0.339
VMD10	99	1.211	0.736	1.148	0.667
VMD11	100	1.078	0.622	1.078	0.591
VMD12	101	0.833	0.495	1.19	0.655
VMD13	102	**0.454**	**0.325**	**0.434**	**0.27**

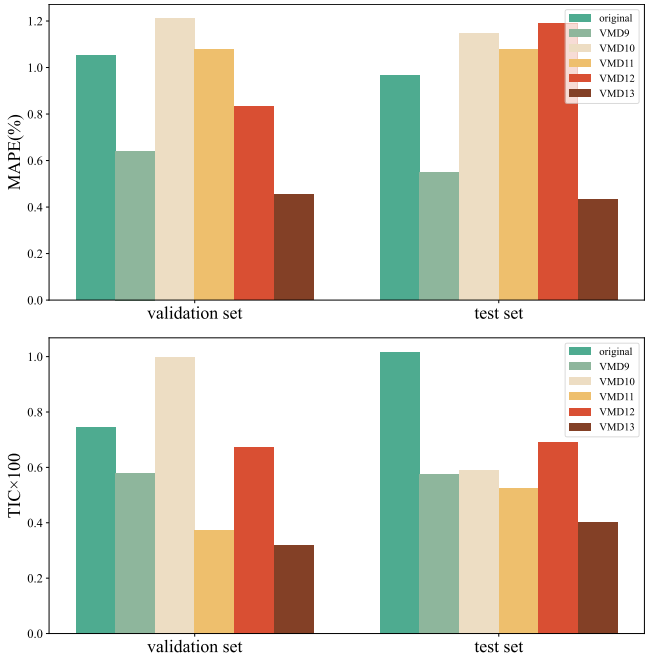


Fig. 13. Prediction accuracy with different numbers of load components.

that valuable features still need to be selected. This further validates the significance of employing the IB, which captures more suitable features for training. Adding IB resulted in a 9% improvement in both MAPE and TIC metrics on the test set compared with the original features.

4.3. Case 3: Effectiveness of VMD-IB

Before conducting this case study, the potential impact of the number of sub-sequences generated by VMD on the model's learning outcomes is considered. During the experiment, we explored the VMD decomposition levels starting from 2. After several experiments, we found that at decomposition levels from 2 to 8, although the prediction efficiency is high, the prediction error is also large; when the decomposition level reaches 13 or above, the prediction efficiency is obviously reduced, while the change of prediction error tends to level off, and the improvement effect is limited. Therefore, in order to strike a balance between model operation efficiency and prediction accuracy, and to study the influence of the number of loaded components on the prediction results, we chose 9 to 13 components for calculation.

According to Table 5, the best prediction is obtained when the number of decompositions is 13. In the test set, there is a maximum improvement of 55% in the MAPE metric and 57% in the TIC metric. However, it is essential to note that each dataset's decomposed components have distinct features, thus necessitating independent analysis of the decomposition number. Based on the findings in Table 6, Fig. 13 is generated to visually represent the prediction accuracy. Including components improves the prediction accuracy, thereby demonstrating

Table 6
Prediction results of VMD-IB.

Number	IB	Dim	Validation set		Test set	
			MAPE (%)	TIC × 100	MAPE (%)	TIC × 100
VMD9	Group1	49	0.798	0.563	0.697	0.447
	Group2	83	0.808	0.591	0.559	0.391
	Group3	97	0.545	0.37	0.444	0.291
	Group4	98	0.7	0.418	0.617	0.363
VMD10	Group1	47	1.049	0.726	1.103	0.653
	Group2	84	0.979	0.622	0.892	0.525
	Group3	98	0.663	0.426	0.543	0.354
	Group4	99	**0.443**	**0.304**	**0.419**	**0.269**
VMD11	Group1	49	0.957	0.682	1.032	0.628
	Group2	85	1.136	0.692	1.073	0.656
	Group3	99	1.117	0.635	1.157	0.637
	Group4	100	0.443	0.655	1.082	0.635
VMD12	Group1	49	1.907	1.048	1.735	0.951
	Group2	85	0.929	0.585	0.899	0.528
	Group3	100	4.12	2.068	3.967	1.965
	Group4	101	0.96	0.552	0.954	0.535
VMD13	Group1	48	1.028	0.654	0.838	0.516
	Group2	85	1.03	0.651	1.012	0.597
	Group3	100	0.603	0.399	0.525	0.325
	Group4	101	1.066	0.595	0.98	0.537

the meaningfulness of utilizing VMD to construct load component features. Although there are some deviations between the results of the validation and test sets, they show a similar trend. Specifically, when better results are achieved on the validation set, the results on the test set also tend to be relatively better. This suggests that selecting parameters based on the smallest TIC value in the validation set is a reasonable criterion. The load component features constructed by VMD-IB, result in multiple datasets, each showing varying prediction accuracies after iterative Boruta screening, as depicted in Table 6.

Table 6 highlights that the VMD10 Group4 configuration achieved the minimum TIC value of 0.304 on the validation set, corresponding to the best prediction results for the selected test set. Regarding the test set, the dataset demonstrates a significant improvement in the MAPE metric of 57% and a TIC metric improvement of 60% compared with the original features. These improvements are notably higher compared to the previous sections, where VMD and IB are used individually, thereby providing further evidence of the effectiveness and superiority of the VMD-IB method in load forecasting.

4.4. Case 4: Effectiveness of CNNA-BiLSTM

In this section, the effectiveness of the combined model is validated through ablation experiments using selected VMD10 Group4 data. The results, presented in Table 7 and Fig. 14, confirm the prediction effectiveness of the CNN, BiLSTM, and CNN-BiLSTM models and their respective incorporation of SAM modules. The findings reveal that the CNN-BiLSTM model outperforms the individual CNN or BiLSTM models regarding to prediction accuracy. This indicates that the combination of CNN, which excels at extracting time-series features, and BiLSTM, which captures longer-distance dependencies, is more advantageous for power load prediction. Moreover, the model with the added SAM module demonstrates higher accuracy compared with the original model, implying that the SAM module enhances the weights of more important features and contributes to improved model fitting. This further supports the notion that the CNNA-BiLSTM model shows superior prediction capability.

Additionally, experiments are conducted to compare the performance of the CNNA-BiLSTM model with other well-known nonlinear intelligent algorithms, namely Support Vector Regression (SVR), AdaBoost, Multi-Layer Perceptron Regression (MLR), Decision Tree Regression (DTR), Backpropagation Neural Network (BP), Gated Recurrent Unit (GRU), and CNNA-LSTM model. These models are evaluated

Table 7
Results of ablation experiments for the combined model.

Model	Original model		Combined with the SAM	
	MAPE (%)	TIC × 100	MAPE (%)	TIC × 100
CNN	0.766	0.451	0.567	0.358
BiLSTM	0.572	0.36	0.455	0.293
CNN-BiLSTM	0.45	0.281	**0.419**	**0.269**

Table 8
Results of different models.

Model	Original data		VMD10 Group4	
	MAPE (%)	TIC × 100	MAPE (%)	TIC × 100
SVR	1.209	0.751	0.928	0.583
AdaBoost	1.154	0.711	0.952	0.614
MLR	2.64	1.565	1.633	1.043
DTR	1.745	1.211	1.634	1.053
BP	1.357	0.798	0.579	0.353
GRU	1.232	0.745	0.626	0.391
CNNA-LSTM	1.069	0.628	0.449	0.334
Proposed model	**0.967**	**0.622**	**0.419**	**0.269**

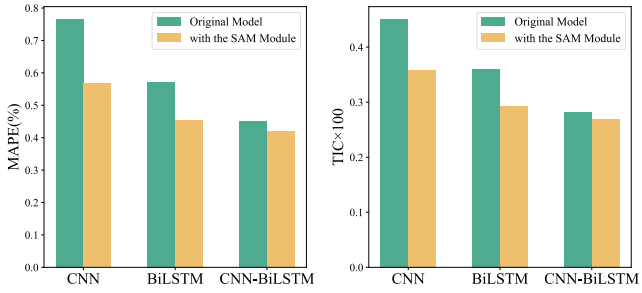


Fig. 14. Results of ablation experiments for the combined model.

and compared with the proposed CNNA-BiLSTM model, and the results are summarized in Table 8.

As shown in the Table 8, the CNNA-BiLSTM model proposed in this paper achieves the best prediction results in both datasets. The robustness of the VMD-IB in model training is assessed by comparing VMD10 Group4 and the original data using the same model. The results indicate that the model trained with VMD10 Group4 for load prediction yields improved forecast performance, with up to 58% enhancement in prediction accuracy. The load prediction results are illustrated in Fig. 15.

4.5. Case 5: Effectiveness of the proposed method over time

This section investigates the influence of seasonality and time factors on the research outcomes. We conducted longitudinal comparison experiments spanning from February 15, 2020, to August 15, 2021, aiming to validate the effectiveness of our proposed method further. These experiments maintained consistency in model settings, sub-dataset time spans, and other relevant factors. The load forecasting tasks' performance is evaluated using a sliding time window approach. For instance, the training set for load forecasting covered the period from November 15, 2020, to May 1, 2021, and the prediction is made for the load from May 9 to May 15, 2021. Meanwhile, the validation set consisted of data from May 2 to May 8, 2021. Each sub-dataset had a duration of six months. The datasets are selected using VMD-IB on the validation set, as shown in the Table 10. Based on the findings in Table 9, our proposed method outperforms LSTM-MTL [33] in terms of MAPE for 10 out of 13 datasets and TIC for 7 out of 13 datasets. Moreover, the average results from comprehensive experiments demonstrate that our proposed method achieves higher prediction accuracy, with MAPE and TIC values of 0.4% and 0.271, respectively.

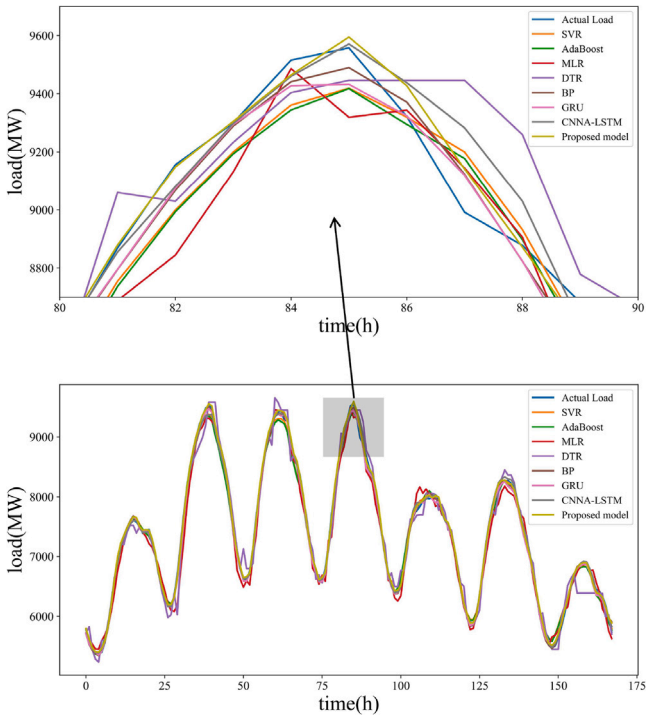


Fig. 15. Prediction results of different model comparisons.

Table 9
Longitudinal comparison experiments across different time periods.

Index	Date	MAPE (%)		TIC × 100	
		LSTM-MTL	Proposed method	LSTM-MTL	Proposed method
1	8.09–8.15,2020	0.467	**0.419**	0.311	**0.269**
2	9.09–9.15,2020	**0.389**	0.439	**0.27**	0.272
3	10.09–10.15,2020	0.406	**0.362**	0.245	**0.249**
4	11.09–11.15,2020	0.427	**0.421**	**0.262**	0.27
5	12.09–12.15,2020	**0.332**	0.459	**0.208**	0.329
6	1.09–1.15,2021	0.511	**0.358**	0.329	**0.235**
7	2.09–2.15,2021	0.437	**0.34**	0.28	**0.216**
8	3.09–3.15,2021	**0.364**	0.391	**0.232**	0.279
9	4.09–4.15,2021	0.428	**0.271**	0.262	**0.188**
10	5.09–5.15,2021	0.501	**0.4**	0.324	**0.278**
11	6.09–6.15,2021	0.53	**0.403**	0.366	**0.282**
12	7.09–7.15,2021	0.598	**0.54**	**0.373**	0.383
13	8.09–8.15,2021	0.538	**0.403**	0.346	**0.272**
	Average	0.456	**0.4**	0.293	**0.271**

Table 10
Selected datasets for longitudinal comparison experiments.

Index	1	2	3	4	5	6	7
Number	VMD10	VMD9	VMD9	VMD11	VMD13	VMD11	VMD12
IB	Group4	Group4	Group5	/	/	/	Group3
Index	8	9	10	11	12	13	
Number	VMD13	VMD9	VMD12	VMD12	VMD9	VMD13	
IB	Group3	/	Group3	Group5	Group3	/	

5. Conclusion

This paper proposes the VMD-IB method and CNNA-BiLSTM model to improve the accuracy of short-term power load forecasting. In this paper, we have made the following contributions: (1) Comparative

experiments involving EMD, EEMD, and CEEMDAN methods for constructing load component features demonstrate that VMD-generated components show clear advantages over the other methods, significantly enhancing load prediction accuracy. Additionally, IB effectively identifies the most valuable features for load prediction through an adaptive iterative strategy, overcoming the limitations of the Boruta method in feature extraction and providing valuable information for model training. By combining the advantages of both methods, the proposed VMD-IB method enables the decomposition of load data into intrinsic pattern functions, facilitating the extraction of key features that significantly influence load prediction. (2) To validate the effectiveness of the proposed model, we conducted ablation experiments and compared it with mainstream models, including SVR, DTR, and GRU. The experimental results demonstrate the advantages of the CNNA-BiLSTM model. (3) Based on this study's extensive experiments and comparative analysis, the proposed method outperforms the state-of-the-art method regarding load forecasting accuracy. The longitudinal comparison experiments reveal a significant improvement of 12% in MAPE and 8% in TIC. These findings confirm the effectiveness and superiority of our method in accurately predicting power system load. Future research directions may involve the application of the method to wind, photovoltaic, and transportation forecasting, as well as exploring alternative deep learning forecasting methods. These endeavors will broaden the method's scope and foster further advancements in short-term load forecasting.

CRediT authorship contribution statement

Jing Xu: Conceptualization, Methodology, Software, Writing – original draft. **Yan Wei:** Formal analysis, Methodology, Validation. **Pan Zeng:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

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Data availability

I have shared my data link in the manuscript.

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